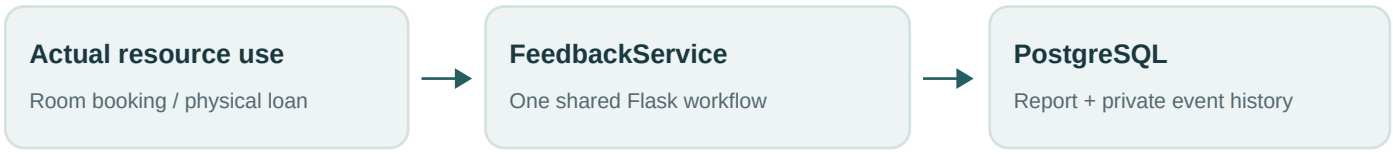
Iteration 48

UC-STU-06 | Resource Feedback and Issue Reporting
University Resource Allocation System - 21 October 2026

Objective and delivered behavior

One optional, private reporting workflow now serves rooms and physical equipment. Students report problems during actual use and give general feedback or improvement suggestions after room completion or verified per-unit return. Skipping feedback never blocks a booking or return.



Existing authentication, notifications, maintenance and waiting lists are reused. C++ / JSON v1 are unchanged.

Area	Implemented behavior
Rooms and groups	Organizer and explicitly registered participants may submit separate private reports. Existing RoomInvitation records support registered students; counts never imply membership. Professor invitations keep their existing workflow.
Equipment	Reports link to an actual reserved physical unit and borrower. Handover proves use; only a verified return enables general feedback. Partial returns are checked separately for each unit.
Responsible staff	Current faculty room manager or equipment custodian reviews, prioritizes, responds and resolves/closes. Open reports support reporter clarification; staff may reopen with an explanation.
Safety and privacy	Reports never automatically block resources or cancel reservations. Existing authorized maintenance remains authoritative. Other participants and unrelated staff cannot read private reports.

Data, lifecycle and integrity

Submitted -> Under Review -> Resolved / Closed; reopening returns to Under Review. Original submission and full actor/time/recipient history remain retained.

Database record	Responsibility
resource_feedback	Immutable actual-use association, reporter, owning faculty and original text; staff assignment, current status/priority/response, timestamps and retry identity.
feedback_event	Private status/action history, complete responses, unique action keys and ordered revisions. Existing notification receipts gain one feedback-event source.

Migration **0048a** adds exactly two tables: **50 application tables**, plus Alembic. Foreign keys, source checks, immutable-association triggers and participant-total checks protect integrity. Row locks, optimistic revisions and idempotency make report/event/notification writes atomic.

Verification and Design Review

Actual executed evidence; no production deployment or Iteration 48 merge.

Verification	Observed result
C++ / CLI	Fresh build: 172 passed, 0 failed; both isolated CSV CLI smokes passed. No engine source or strategy changed.
Full Python suite	400 run: 399 passed, 0 failed, 1 skipped on Windows. Uses real isolated PostgreSQL schemas and the authoritative compiled engine.
Focused follow-up	31 feedback + 18 room + 49 waiting-list tests passed after hardening. Two regressions were added after the full run began; the final suite has 402 cases.
Database	Fresh upgrades, drift, safe downgrade/re-upgrade and destructive-downgrade refusal passed. Real dump/restore and restart passed.
Existing services	All five strategy parity comparisons, appointments, room/equipment, calendar, waiting lists and CSV compatibility passed.
Browser acceptance	Real forms, decisions, private group reports, maintenance and verified-return feedback passed; desktop 1440px, tablet 768px, mobile 390px.
Preservation / skip	Operational database and four preexisting CSV outputs remained unchanged. POSIX execute-permission testing is skipped on Windows; no Linux result is claimed here.

Concrete end-to-end acceptance

An isolated browser workflow verified four reports, 13 history events, participant privacy (HTTP 403) and custodian-verified return. The room stayed confirmed; damaged inventory was not made usable. Notification receipts were unique. Test schemas and browsers were cleaned.

Final review improvements

Maintenance free-text notes are not redistributed between reporters: shared history contains only availability facts, while full notes remain in the original authorized record. Room aggregate validation includes every registered participant in the declared attendee total. Maximum-length descriptions, responses and history wrap within a 390px mobile viewport.

A live PostgreSQL regression reproduced parent/user and authority/user lock inversions. Parent-first locking and FK-compatible NO KEY UPDATE serialization resolve both; duplicate submissions still serialize correctly.

A group waiting-list regression reproduced incorrect professor-only participant validation. Shared participant partitioning now preserves registered students through pre-approval and automatic assignment.

An obsolete staff-configuration denial assertion was replaced by scoped-access and stronger ownership/policy/other-faculty rejection checks; the full suite was rerun.

Architecture decisions and alternatives

Reused application services, transactions, query-scoped permissions and notifications. Separate reporting systems, booking-audience broadcasts and new allocation algorithms were rejected as duplication, privacy risk and unnecessary scope. No new framework or decorative design pattern was added.

Operational limits and handover

Apply migration 0048a before use; tests did not migrate operational data. Inboxes display 200 filtered records; history is retained. No attachments, ratings, anonymity or emergency guarantee. Reports retain their original-use faculty; cross-faculty case transfer is not automated. Remote CI is verified separately on the draft PR.

Reference: docs/architecture/resource-feedback-design.md; feedback-er.mmd / .svg; database physical catalog; UC-STU-06 requirements and approved use case.